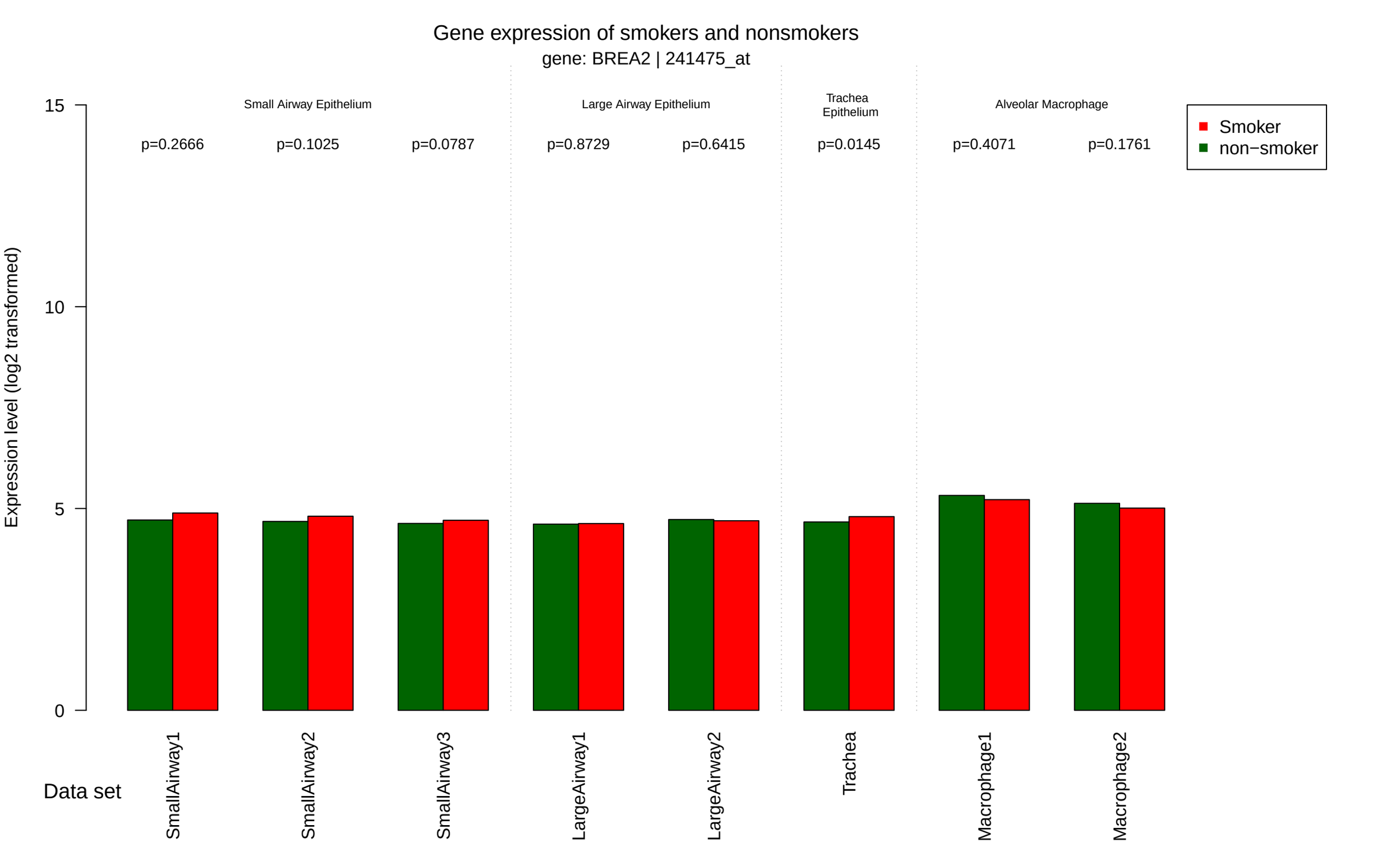
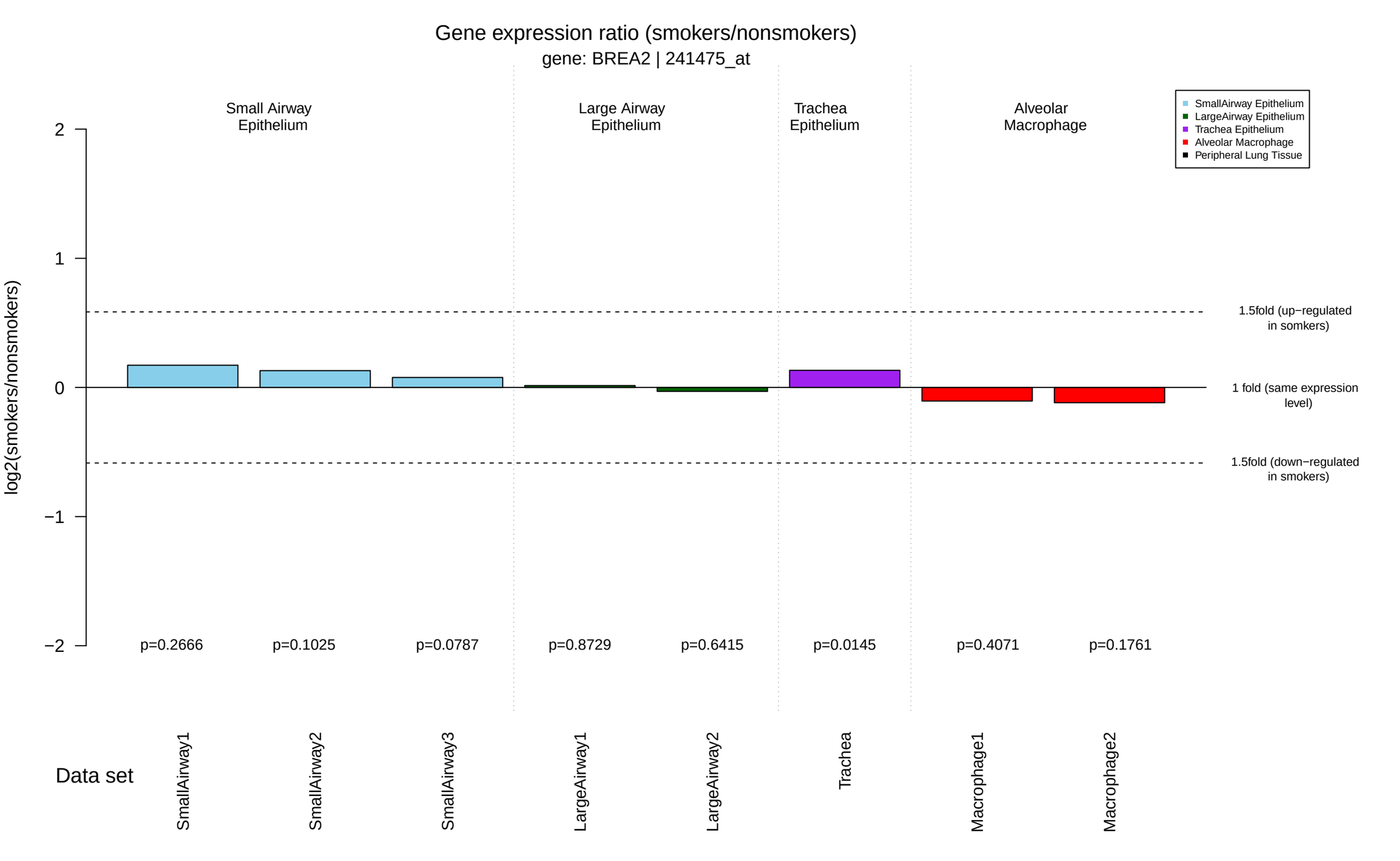
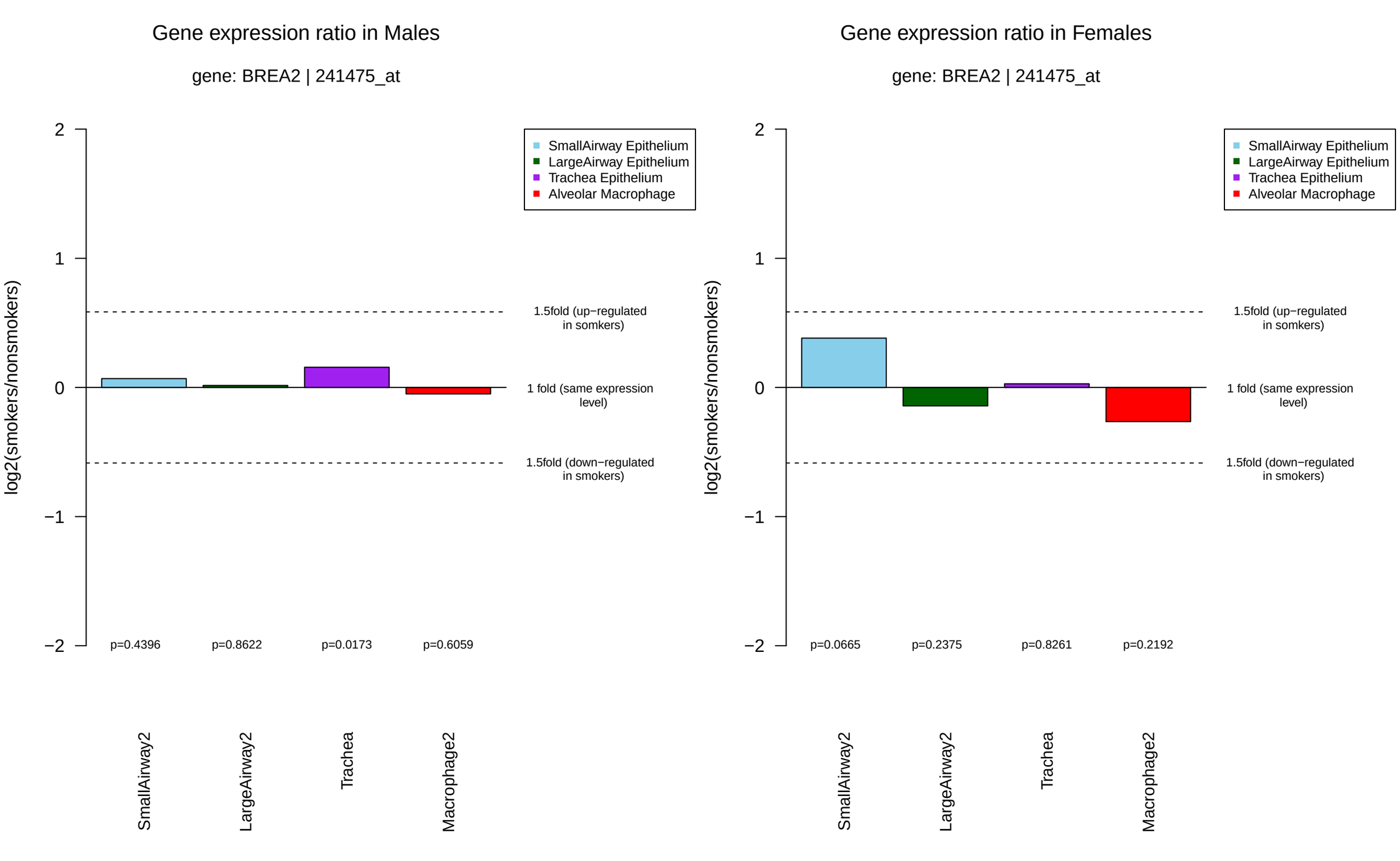
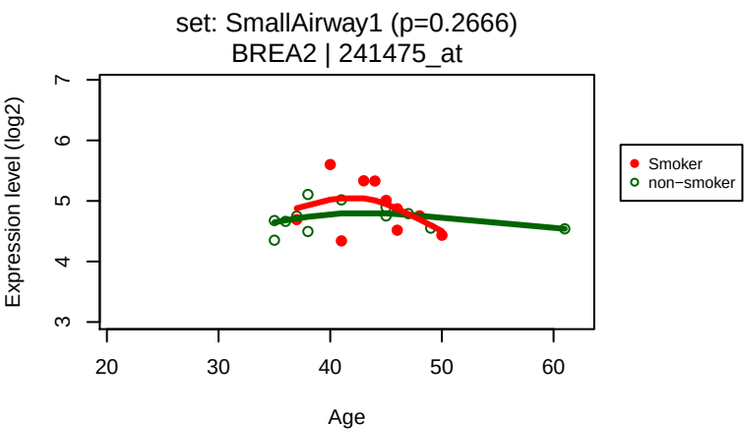
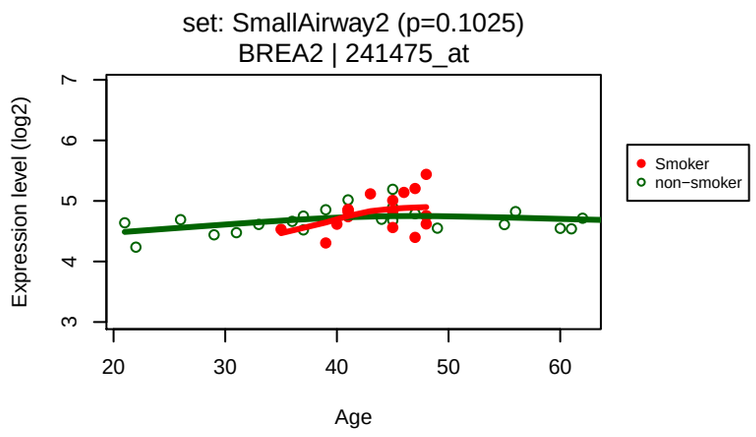
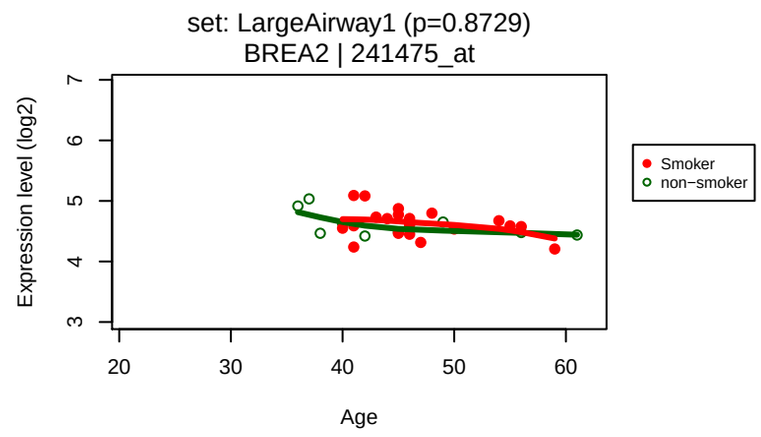

Gene expression of smokers and nonsmokers
gene: BREA2 | 241475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2666
p=0.1025
p=0.0787
p=0.8729
p=0.6415
p=0.0145
p=0.4071
p=0.1761
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

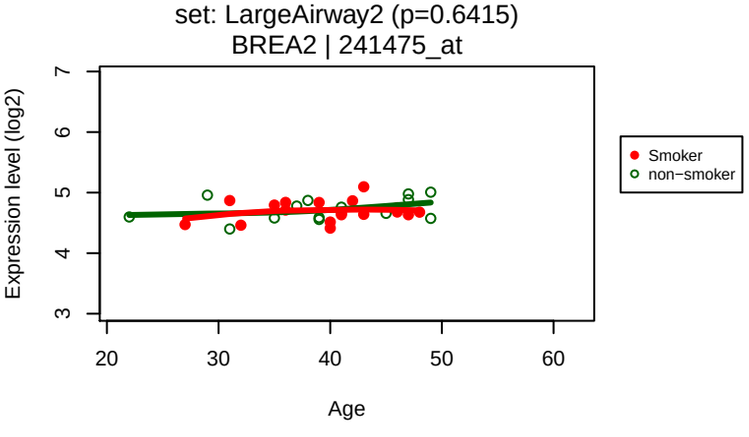

Gene expression ratio (smokers/nonsmokers)
gene: BREA2 | 241475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2666
p=0.1025
p=0.0787
p=0.8729
p=0.6415
p=0.0145
p=0.4071
p=0.1761
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

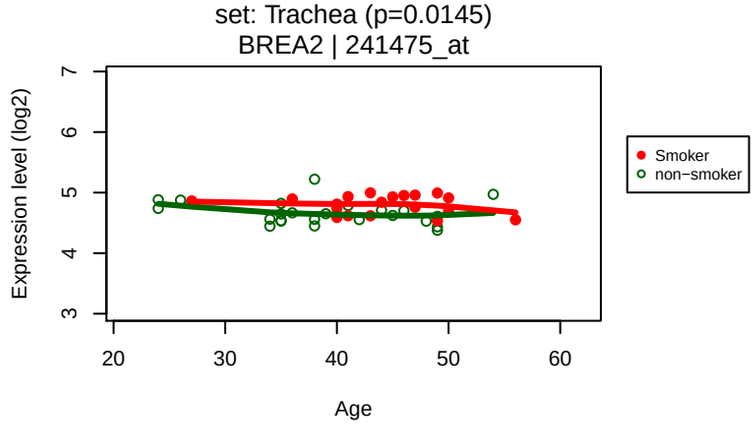

Gene expression ratio in Males
gene: BREA2 | 241475_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4396
p=0.8622
p=0.0173
p=0.6059
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BREA2 | 241475_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0665
p=0.2375
p=0.8261
p=0.2192
SmallAirway2
LargeAirway2
Trachea
Macrophage2

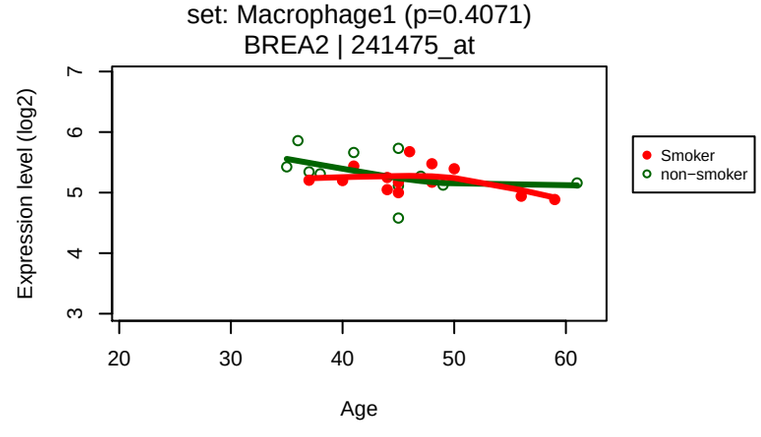

set: SmallAirway1 (p=0.2666)
BREA2 | 241475_at
Expression level (log2)
Age
Smoker
non−smoker

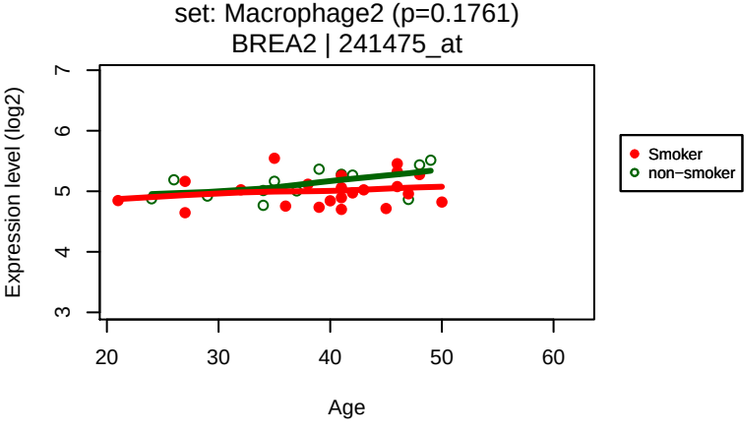

set: SmallAirway2 (p=0.1025)
BREA2 | 241475_at
Expression level (log2)
Age
Smoker
non−smoker

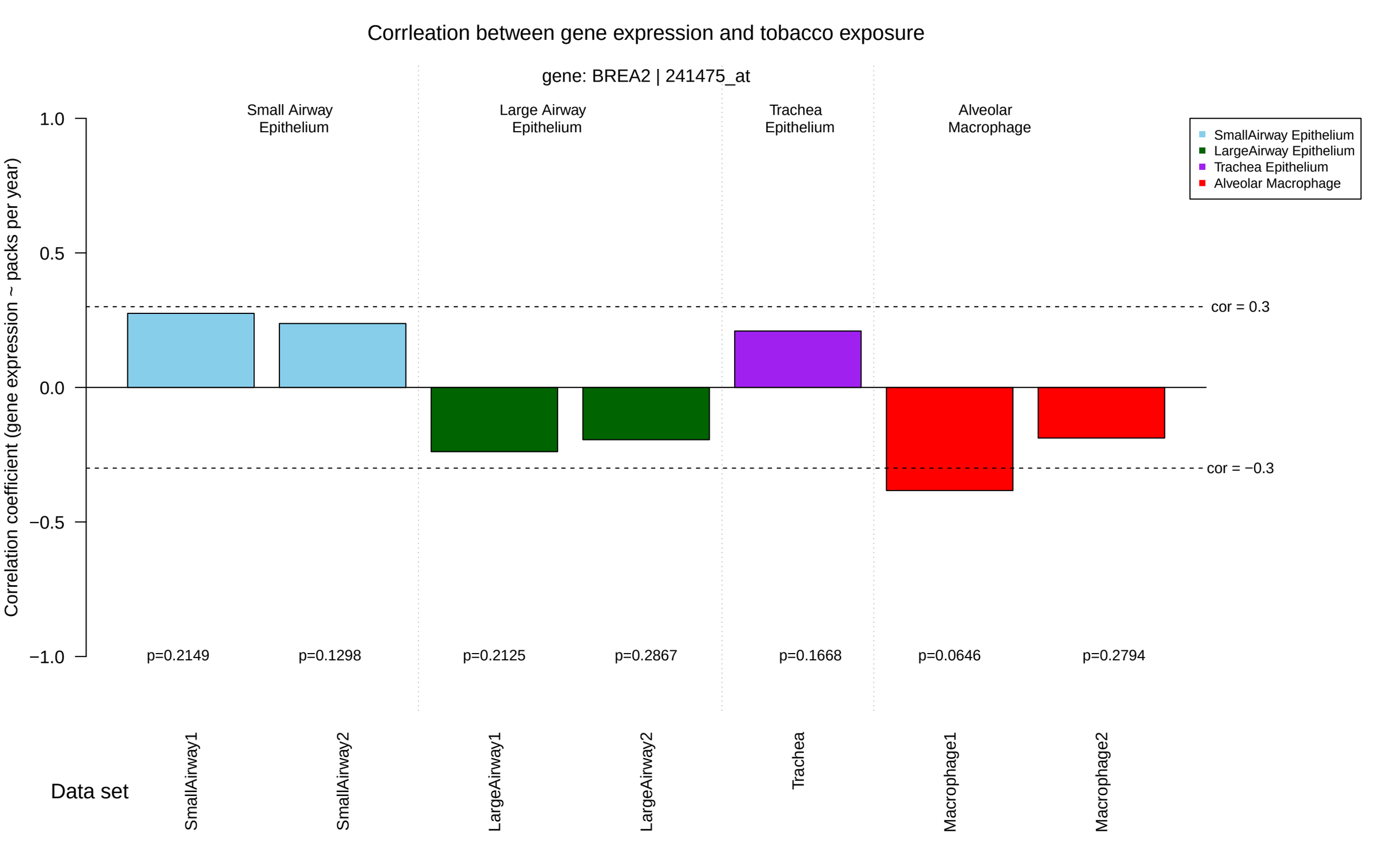

set: LargeAirway1 (p=0.8729)
BREA2 | 241475_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6415)
BREA2 | 241475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0145)
BREA2 | 241475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4071)
BREA2 | 241475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1761)
BREA2 | 241475_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BREA2 | 241475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2149
p=0.1298
p=0.2125
p=0.2867
p=0.1668
p=0.0646
p=0.2794
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2